

## SAFETY DATA SHEET

Creation Date 13-Oct-2009

Revision Date 28-Mar-2024

Revision Number 3

### 1. Identification

**Product Name** Ethyl acetate

**Cat No. :** 39177

**CAS No** 141-78-6  
**Synonyms** Acetic acid ethyl ester

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
**Emergency Number US:**001-201-796-7100 / **Europe:** +32 14 57 52 99  
**CHEMTREC** Tel. No. **US:**001-800-424-9300 / **Europe:**001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |            |
|--------------------------------------------------|------------|
| Flammable liquids                                | Category 2 |
| Serious Eye Damage/Eye Irritation                | Category 2 |
| Specific target organ toxicity (single exposure) | Category 3 |
| Target Organs - Central nervous system (CNS).    |            |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Highly flammable liquid and vapor  
Causes serious eye irritation  
May cause drowsiness or dizziness

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not breathe dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge  
Wear protective gloves/protective clothing/eye protection/face protection  
Keep cool

**Response**

Get medical attention/advice if you feel unwell

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
Call a POISON CENTER or doctor/physician if you feel unwell

**Skin**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
If eye irritation persists: Get medical advice/attention

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Repeated exposure may cause skin dryness or cracking

### 3. Composition/Information on Ingredients

| Component     | CAS No   | Weight % |
|---------------|----------|----------|
| Ethyl acetate | 141-78-6 | <=100    |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

|                                            |                                                                                                                                                                                           |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                              |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                                                              |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. May cause central nervous system depression: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                     |

## 5. Fire-fighting measures

|                                         |                                                                                             |
|-----------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam.       |
| <b>Unsuitable Extinguishing Media</b>   | Water may be ineffective, Do not use a solid water stream as it may scatter and spread fire |
| <b>Flash Point</b>                      | -4 °C / 24.8 °F                                                                             |
| <b>Method -</b>                         | CC (closed cup)                                                                             |
| <b>Autoignition Temperature</b>         | 427 °C / 800.6 °F                                                                           |
| <b>Explosion Limits</b>                 |                                                                                             |
| <b>Upper</b>                            | 11.5 vol %                                                                                  |
| <b>Lower</b>                            | 2.0 vol %                                                                                   |
| <b>Oxidizing Properties</b>             | Not oxidising                                                                               |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                    |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                    |

### Specific Hazards Arising from the Chemical

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 2             | 3                   | 0                  | N/A                     |

## 6. Accidental release measures

|                                  |                                                                                                    |
|----------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Use personal protective equipment as required. Ensure adequate ventilation.                        |
| <b>Environmental Precautions</b> | Should not be released into the environment. See Section 12 for additional Ecological Information. |

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

|                 |                                                                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.                     |
| <b>Storage.</b> | Flammables area. Keep away from heat, sparks and flame. Keep container tightly closed in a dry and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong |

acids. Amines. Peroxides.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component     | ACGIH TLV    | OSHA PEL                                                                                                       | NIOSH                                                         | Mexico OEL (TWA) |
|---------------|--------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------------|
| Ethyl acetate | TWA: 400 ppm | (Vacated) TWA: 400 ppm<br>(Vacated) TWA: 1400 mg/m <sup>3</sup><br>TWA: 400 ppm<br>TWA: 1400 mg/m <sup>3</sup> | IDLH: 2000 ppm<br>TWA: 400 ppm<br>TWA: 1400 mg/m <sup>3</sup> | TWA: 400 ppm     |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

### Personal Protective Equipment

#### Eye/face Protection

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

#### Respiratory Protection

No protective equipment is needed under normal use conditions.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                                              |
|----------------------------------------|----------------------------------------------|
| Physical State                         | Liquid                                       |
| Appearance                             | Colorless                                    |
| Odor                                   | sweet                                        |
| Odor Threshold                         | 50 ppm                                       |
| pH                                     | No information available                     |
| Melting Point/Range                    | -83.5 °C / -118.3 °F                         |
| Boiling Point/Range                    | 75 - 78 °C / 167 - 172.4 °F                  |
| Flash Point                            | -4 °C / 24.8 °F                              |
| Method -                               | CC (closed cup)                              |
| Evaporation Rate                       | 6.2                                          |
| Flammability (solid,gas)               | Not applicable                               |
| Flammability or explosive limits       |                                              |
| Upper                                  | 11.5 vol %                                   |
| Lower                                  | 2.0 vol %                                    |
| Vapor Pressure                         | 103 mbar @ 20°C                              |
| Vapor Density                          | 3.04                                         |
| Specific Gravity                       | 0.902                                        |
| Solubility                             | Slightly soluble in water                    |
| Partition coefficient; n-octanol/water | No data available                            |
| Autoignition Temperature               | 427 °C / 800.6 °F                            |
| Decomposition Temperature              | No information available                     |
| Viscosity                              | 0.45 cP @ 20 °C                              |
| Molecular Formula                      | C <sub>4</sub> H <sub>8</sub> O <sub>2</sub> |
| Molecular Weight                       | 88.11                                        |
| Surface tension                        | 24 mN/m @ 20°C                               |

## 10. Stability and reactivity

|                                         |                                                                                          |
|-----------------------------------------|------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                               |
| <b>Stability</b>                        | Stable under normal conditions.                                                          |
| <b>Conditions to Avoid</b>              | Incompatible products. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong acids, Amines, Peroxides                                 |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                  |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                 |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                            |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component     | LD50 Oral            | LD50 Dermal                                       | LC50 Inhalation    |
|---------------|----------------------|---------------------------------------------------|--------------------|
| Ethyl acetate | 10,200 mg/kg ( Rat ) | > 20 mL/kg ( Rabbit )<br>> 18000 mg/kg ( Rabbit ) | 58 mg/l (rat; 8 h) |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | Irritating to eyes                                                                       |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component     | CAS No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------|----------|------------|------------|------------|------------|------------|
| Ethyl acetate | 141-78-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Central nervous system (CNS)  
**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** May cause central nervous system depression: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

**Ecotoxicity**

Do not empty into drains.

| Component     | Freshwater Algae     | Freshwater Fish                                                      | Microtox                                                                                             | Water Flea          |
|---------------|----------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------|
| Ethyl acetate | EC50 = 3300 mg/L/48h | Fathead minnow: LC50: 230 mg/l/ 96h<br>Gold orfe: LC50: 270 mg/L/48h | EC50 = 1180 mg/L 5 min<br>EC50 = 1500 mg/L 15 min<br>EC50 = 5870 mg/L 15 min<br>EC50 = 7400 mg/L 2 h | EC50 = 717 mg/L/48h |

**Persistence and Degradability** Persistence is unlikely based on information available.**Bioaccumulation/ Accumulation** No information available.**Mobility** Will likely be mobile in the environment due to its volatility.

| Component     | log Pow |
|---------------|---------|
| Ethyl acetate | 0.73    |

**13. Disposal considerations**

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component                | RCRA - U Series Wastes | RCRA - P Series Wastes |
|--------------------------|------------------------|------------------------|
| Ethyl acetate - 141-78-6 | U112                   | -                      |

**14. Transport information****DOT**

UN-No UN1173  
 Proper Shipping Name ETHYL ACETATE  
 Hazard Class 3  
 Packing Group II

**TDG**

UN-No UN1173  
 Proper Shipping Name ETHYL ACETATE  
 Hazard Class 3  
 Packing Group II

**IATA**

UN-No UN1173  
 Proper Shipping Name ETHYL ACETATE  
 Hazard Class 3  
 Packing Group II

**IMDG/IMO**

UN-No UN1173  
 Proper Shipping Name ETHYL ACETATE  
 Hazard Class 3  
 Packing Group II

**15. Regulatory information****United States of America Inventory**

| Component     | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|---------------|----------|------|-----------------------------------------------|-----------------------------|
| Ethyl acetate | 141-78-6 | X    | ACTIVE                                        | -                           |

**Legend:****TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b) - Notices of Export**

Not applicable

**International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component     | CAS No   | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|---------------|----------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Ethyl acetate | 141-78-6 | X   | -   | 205-500-4 | X     | X    | X    | X    | X     | KE-00047 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

Not applicable

**Clean Air Act**

Not applicable

**OSHA - Occupational Safety and Health Administration**

Not applicable

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component     | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances RQs | SARA Reportable Quantity (RQ) |
|---------------|--------------------------|-------------------------------------------|-------------------------------|
| Ethyl acetate | 5000 lb                  | -                                         | 5000 lb<br>2270 kg            |

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component     | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|---------------|---------------|------------|--------------|----------|--------------|
| Ethyl acetate | X             | X          | X            | -        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations**

Mexico - Grade

Serious risk, Grade 3

**Authorisation/Restrictions according to EU REACH**

| Component     | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------|----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Ethyl acetate | 141-78-6 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

**REACH links**
<https://echa.europa.eu/substances-restricted-under-reach>
**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component     | CAS No   | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------|----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Ethyl acetate | 141-78-6 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component     | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Ethyl acetate | 141-78-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Annex I - Y42                      |

**16. Other information****Prepared By**

Health, Safety and Environmental Department  
 Email: chem.techinfo@thermofisher.com  
 www.thermofisher.com

**Creation Date**

13-Oct-2009

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28-Mar-2024

**Print Date**

28-Mar-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**